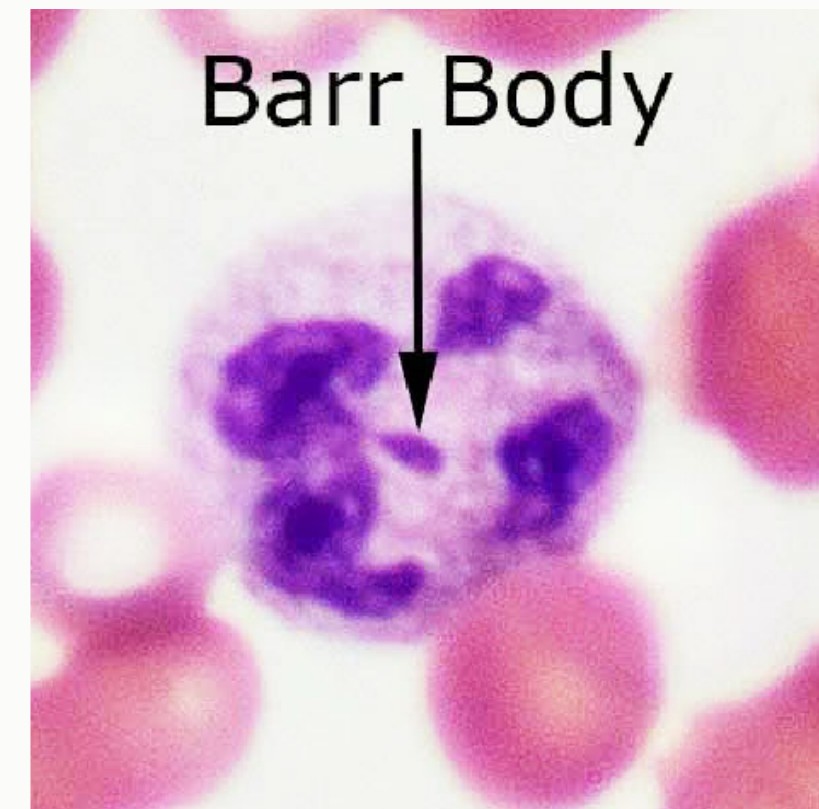


SEX CHROMATIN:

BARR BODIES & DRUMSTICK

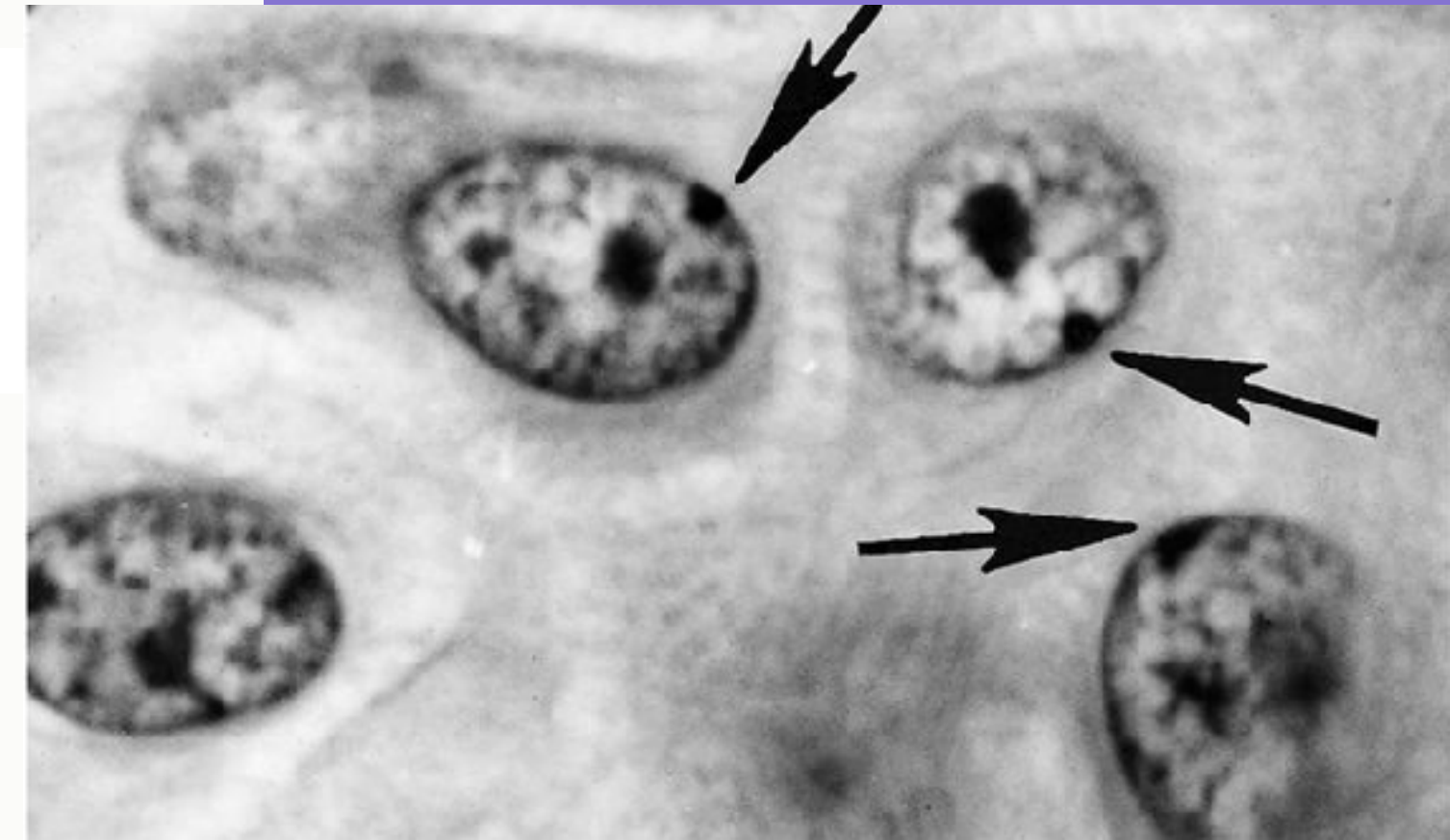
Adan Alnaamneh



sex chromatin mass

The term “**sex chromatin mass**” denotes a body constituting a sex chromosome, or part thereof, which is visible in interphase nucleus.

Barr bodies represent the inactive X chromosome and are normally found only in female somatic cells.



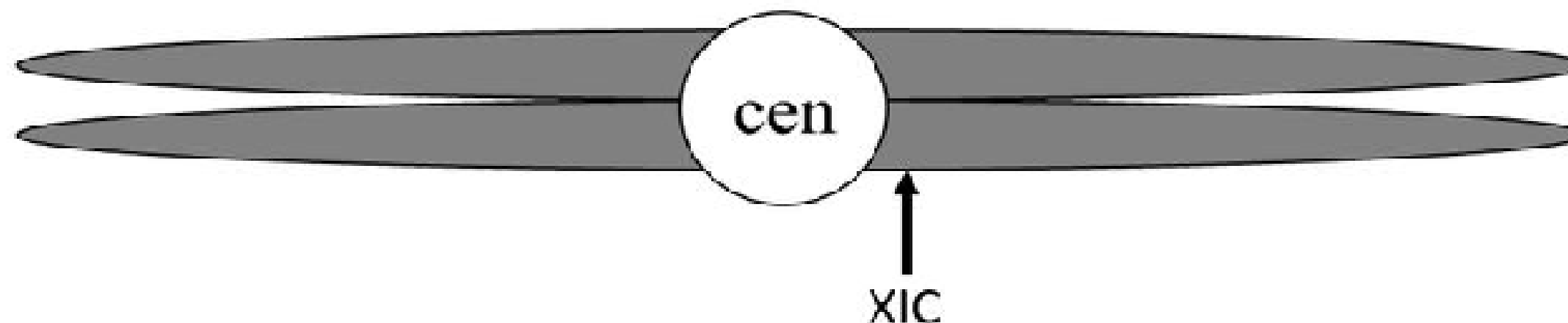
X-Inactivation

- In cells with more than two X chromosomes, only one X remains genetically active and all the others become inactivated.
- In some cells the paternal allele is expressed
- In other cells the maternal allele is expressed
- In XXX and XXXX females and XXY males only 1 X is activated in any given cell the rest are inactivated.
- In human embryos, sex chromatin bodies have been observed by the 16th day of gestation.



Mechanism of X-chromosome Inactivation

- A region of X chromosome near the centromere called the **X-inactivation center (XIC)** is the control unit.
- Contains 12 genes
 - 7 genes code for proteins
 - 5 genes code for untranslated RNA



- This region contains two genes actively involved in inactivating an X chromosome
 - Xist (X-inactive specific transcript)
 - Tsix

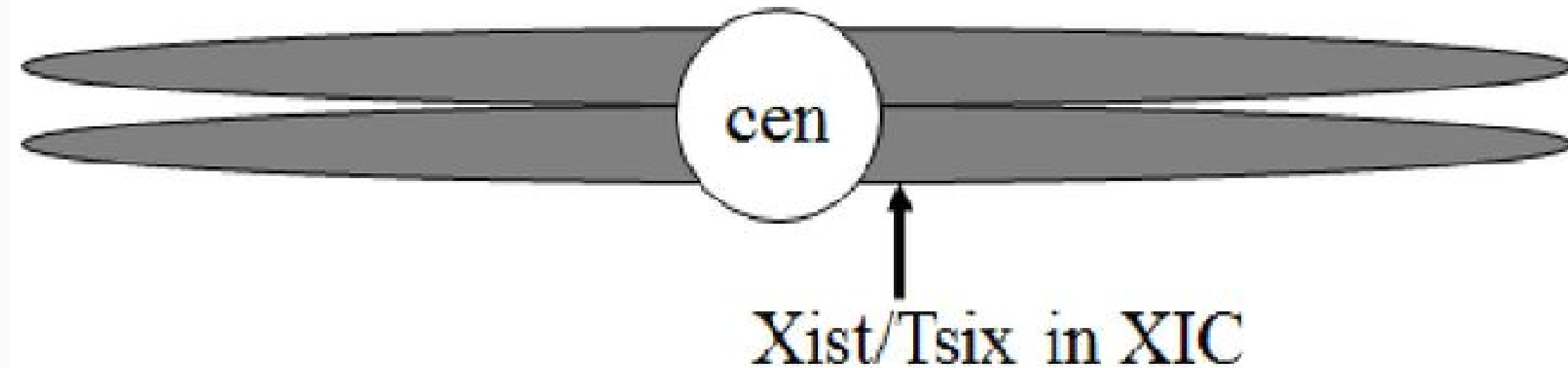
Antagonistic Roles:

Active X (Xa):

↓ Xist expression
↑ Tsix expression

Inactive X (Xi):

↑ Xist expression
↓ Tsix expression

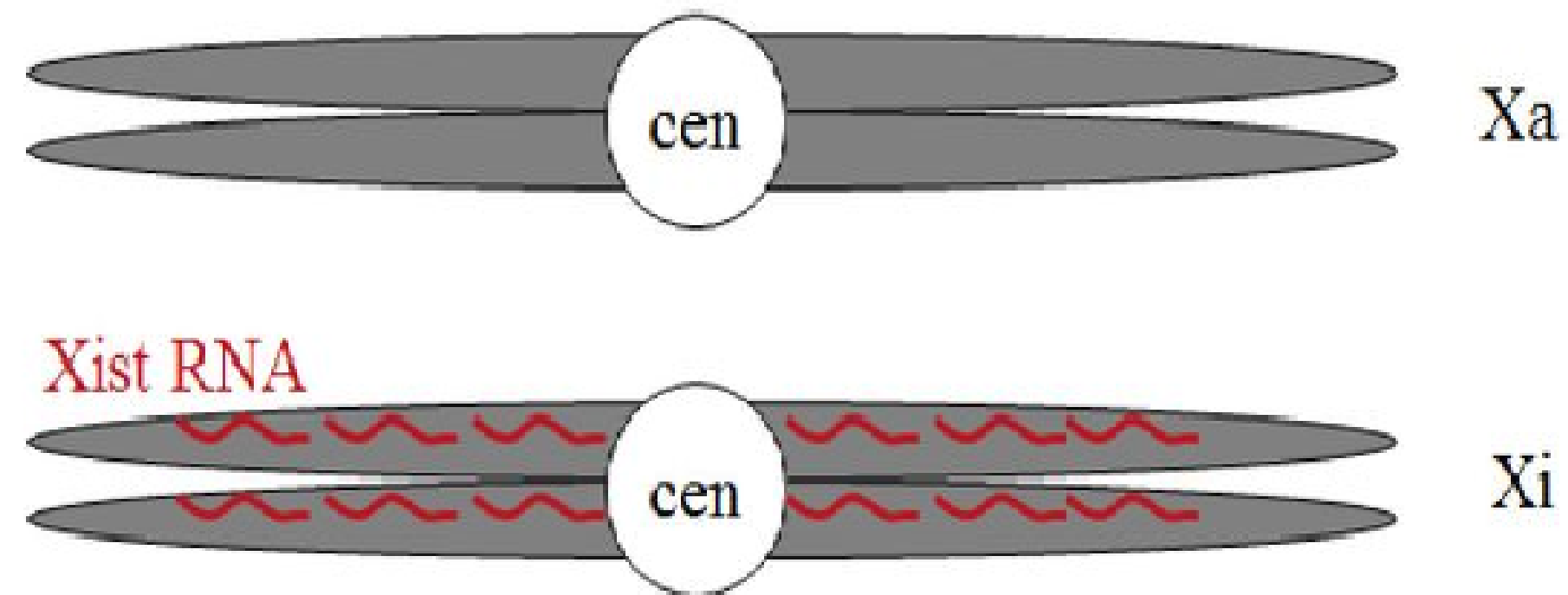


Xist Gene:

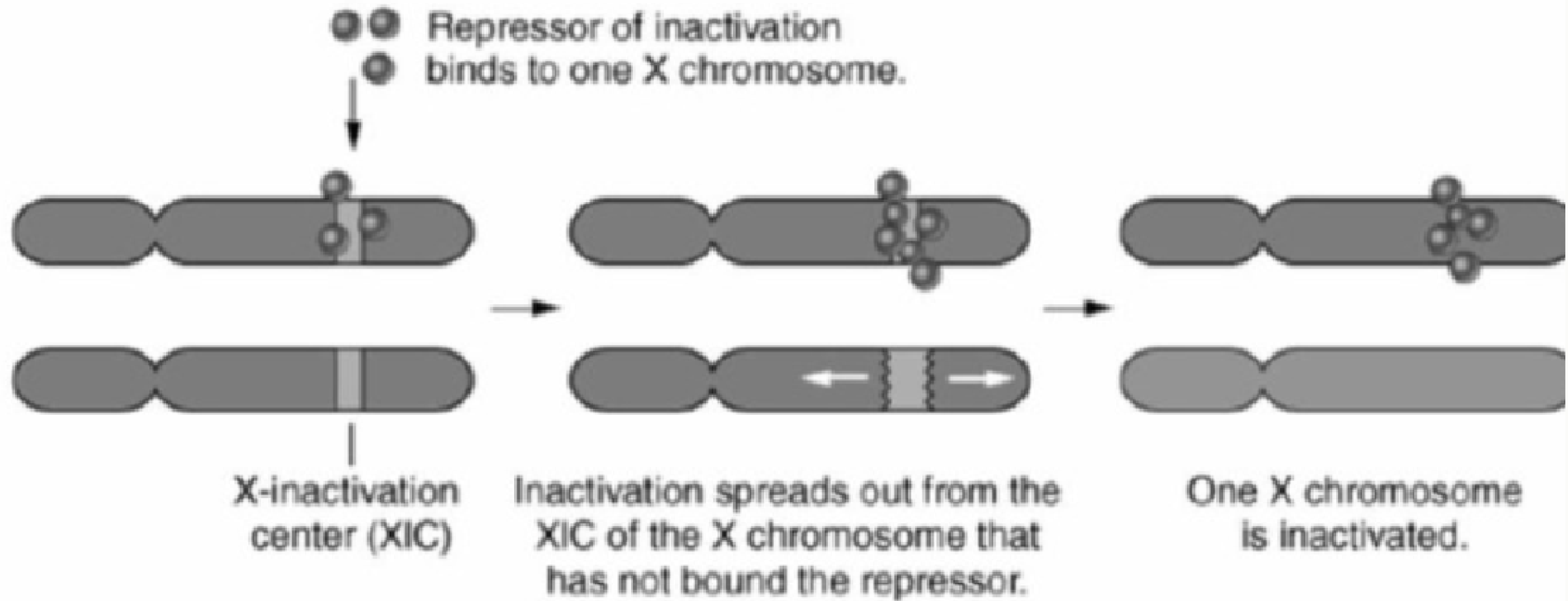
The Xist gene is the only gene which is expressed from the Xi but not from the Xa.

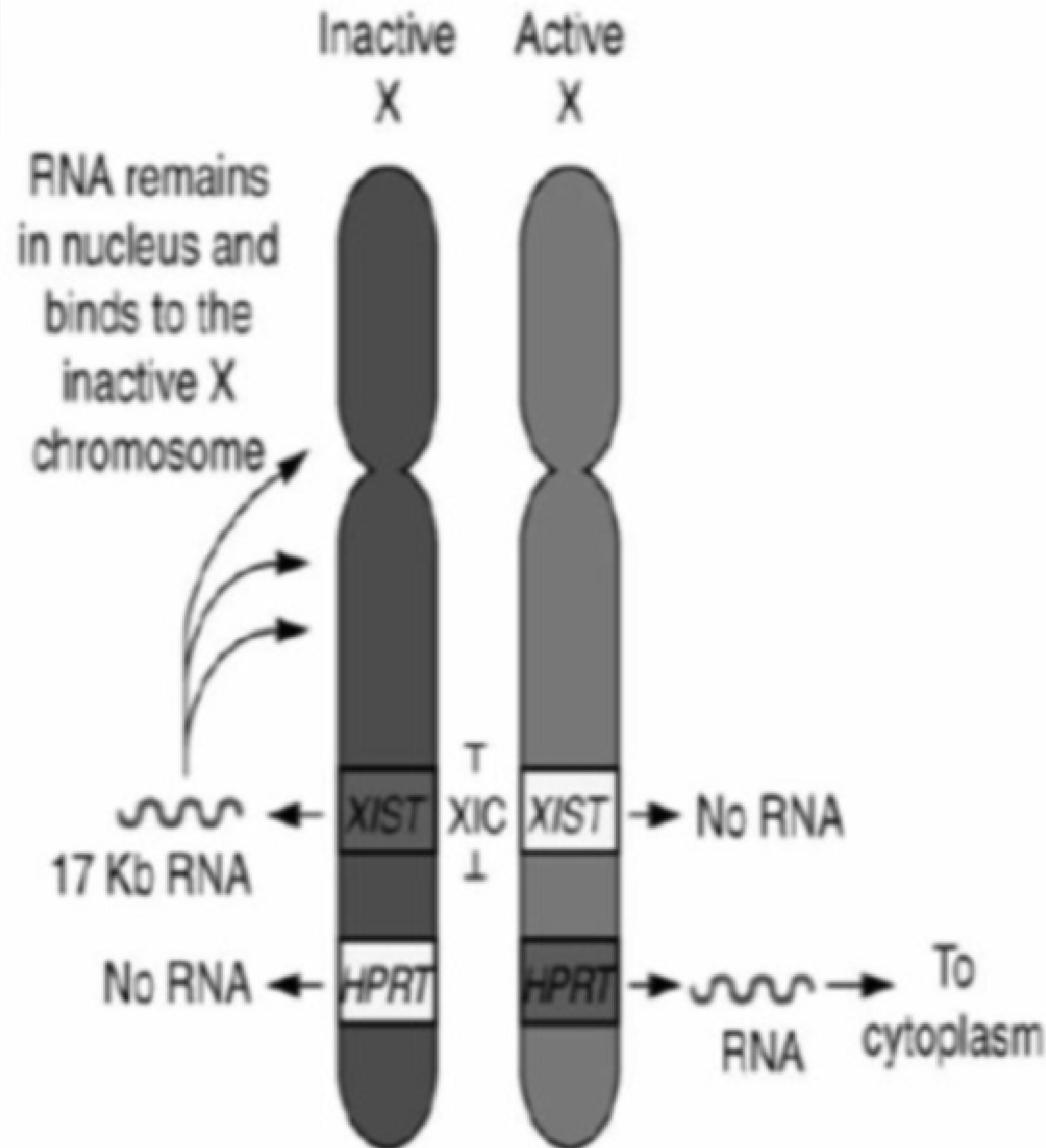
Encodes a large RNA molecule

Coats Xi from the XIC near the centromere outward along the X chromosome



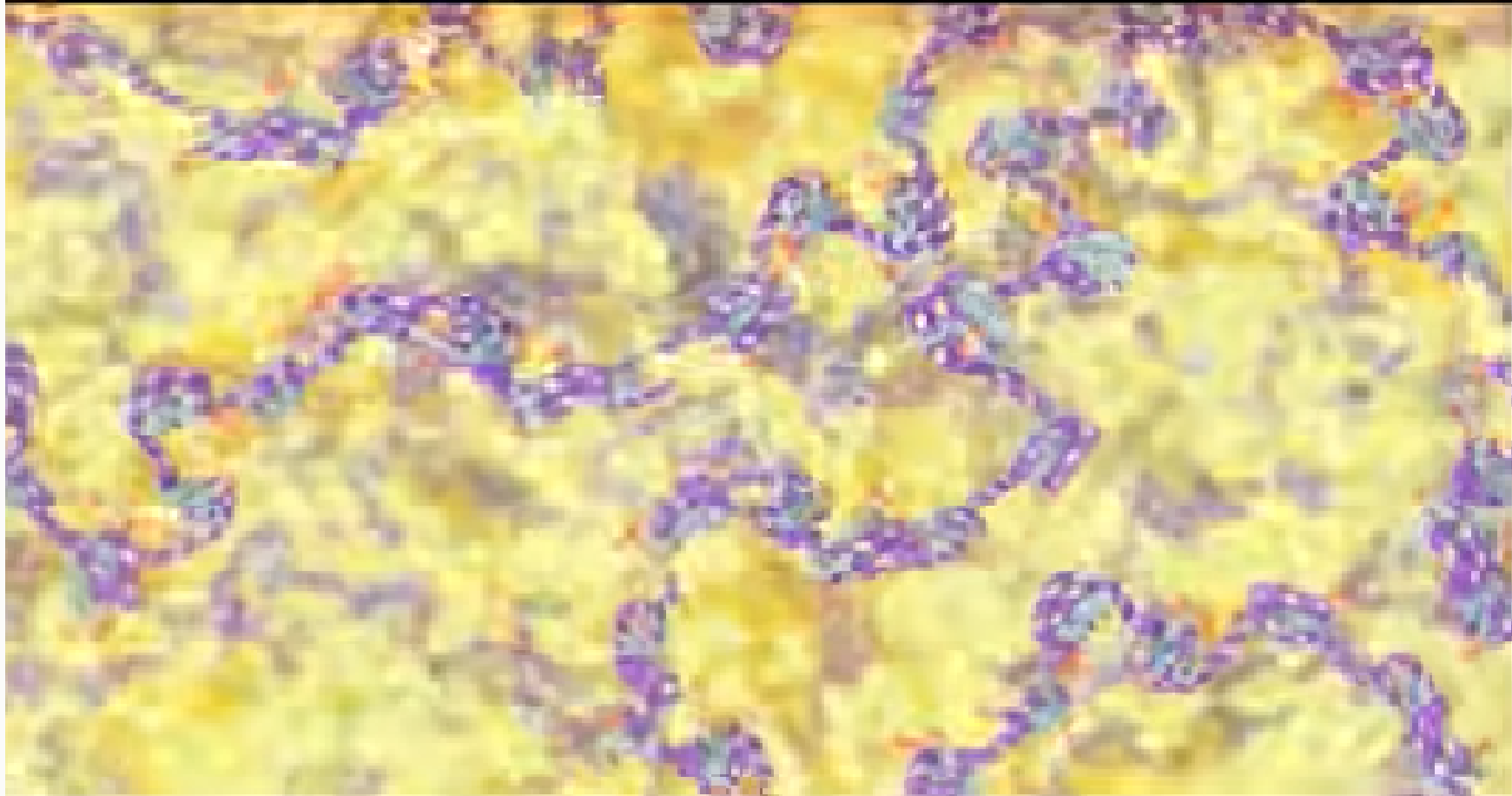
Mechanism of X-inactivation



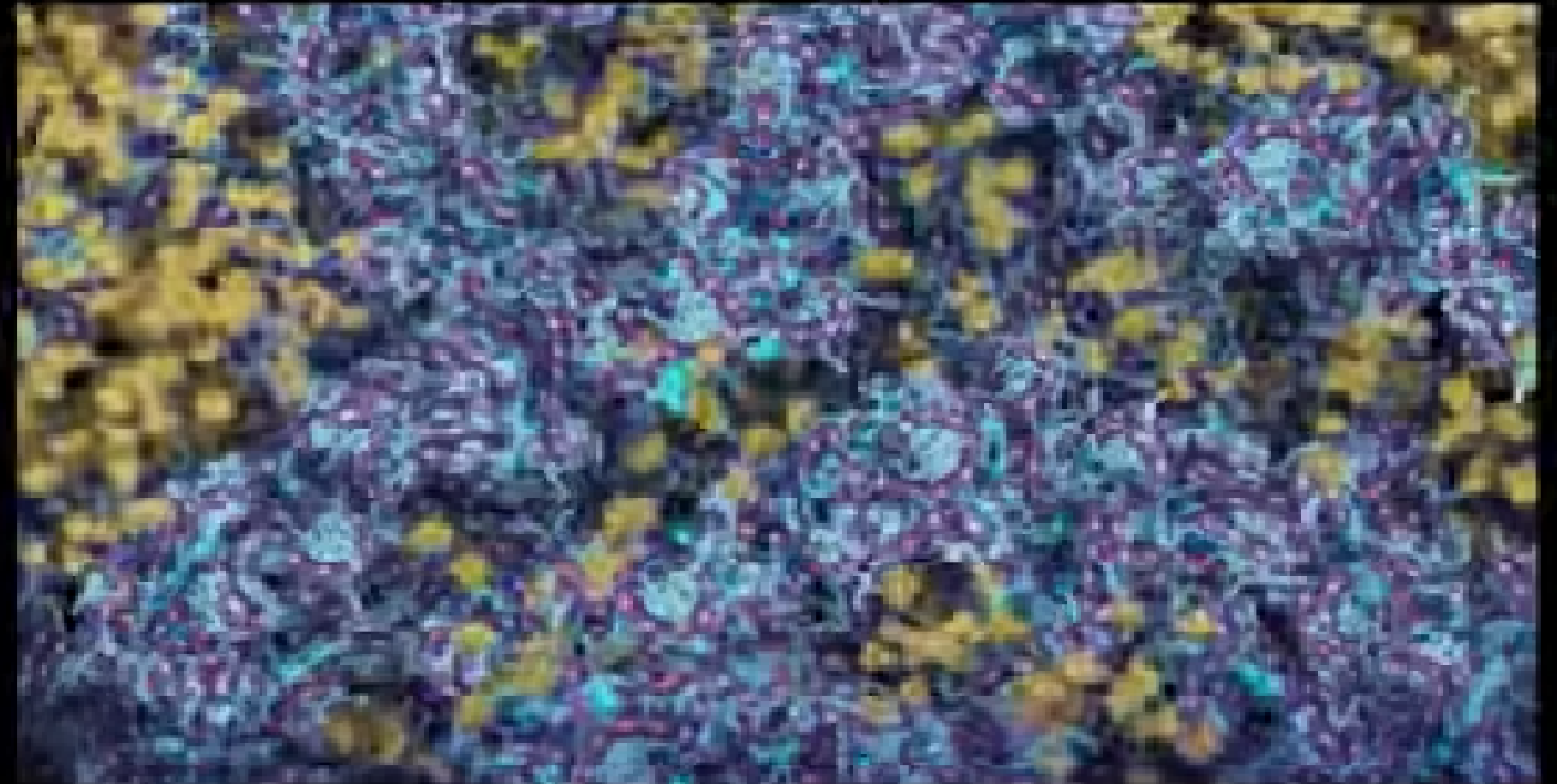


- RNA product from this gene does not leave the nucleus, so it is not translated into protein
- Instead RNA acts as a “cage” that envelopes the X destined to be inactivated
- Further transcription of this RNA on the active X is repressed by methylation of the gene’s promoter region; also true for X in males

X chromosomes



Active
Genes on



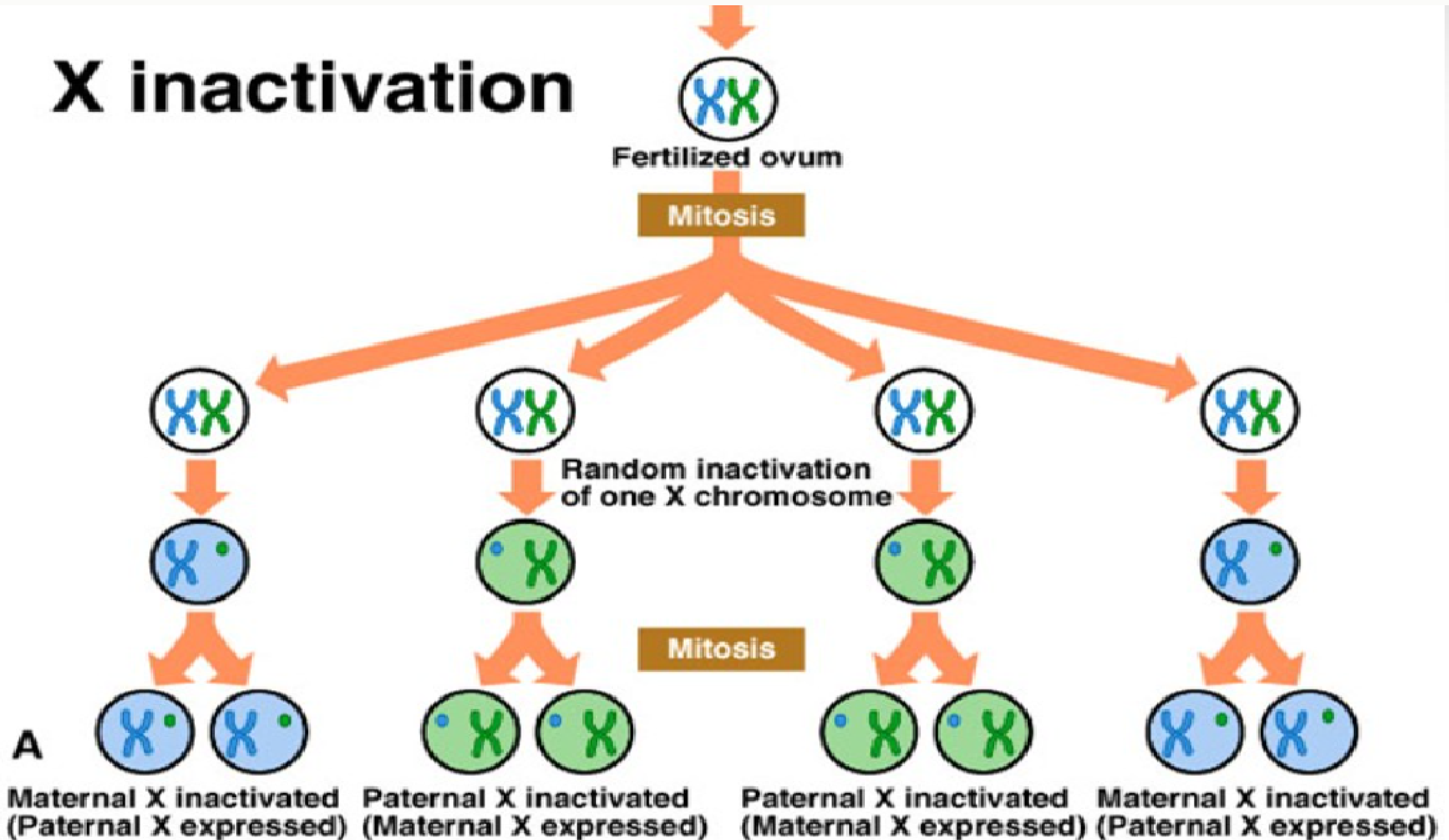
Inactive
Genes off

X-inactivation ensures that females, like males, have one functional copy of the X chromosome in each body cell.

Dosage compensation

- In cells with more than two X chromosomes, **only one X remains genetically active and all the others become inactivated.**
- The process is **independent from cell to cell.**
- A maternal or paternal X is **randomly** chosen to be inactivated.

X inactivation

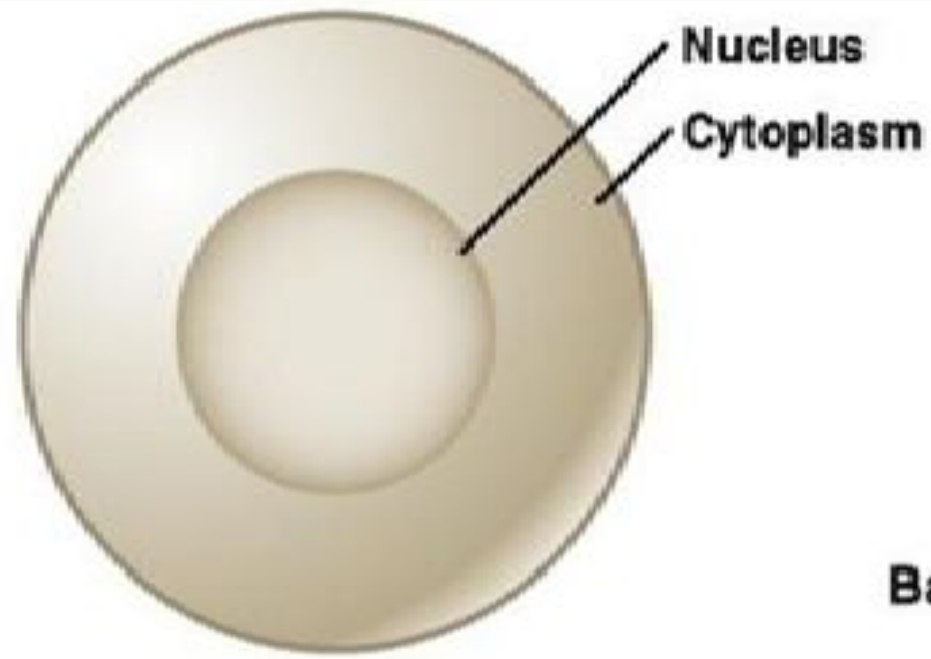


X-inactivation facts about calico cats

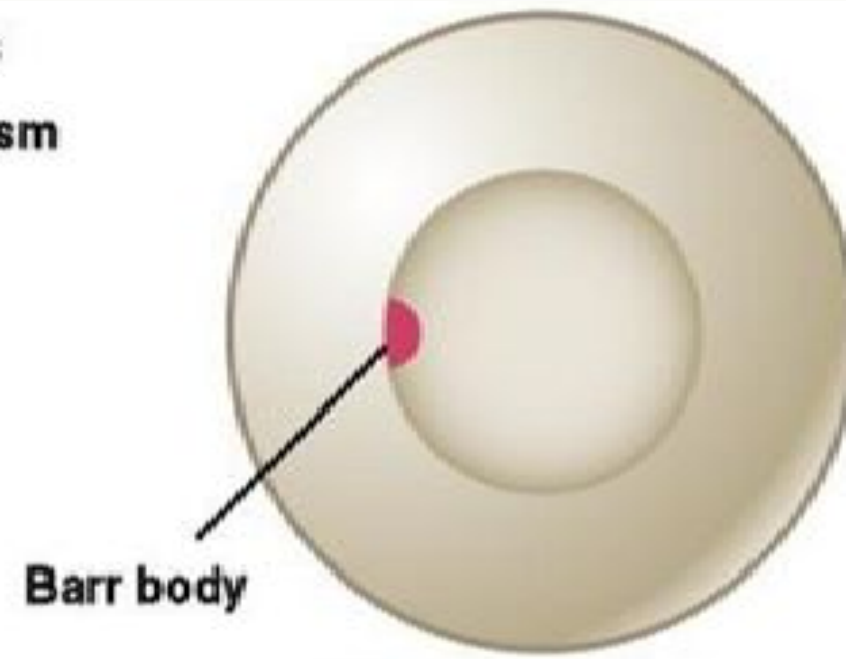


Calico Cats

- only females can be heterozygous for coat color gene on the X chromosome
- one allele codes for orange; the other allele codes for black
- 1/2 of cells have inactivated the black allele--results in patches of orange fur
- 1/2 of cells have inactivated the orange allele--results in patches of black fur
- males are either black or orange, because they are hemizygous for the X-linked color gene



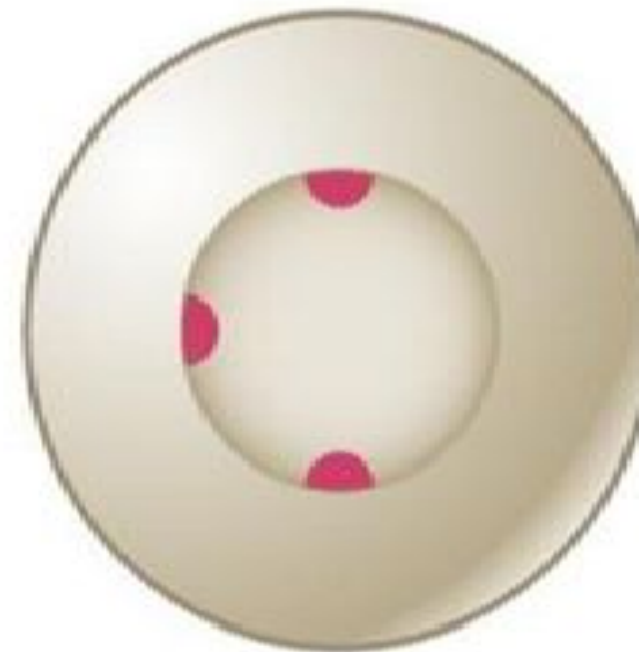
46, XY
45, X
($N - 1 = 0$)



46, XX
47, XXY
($N - 1 = 1$)



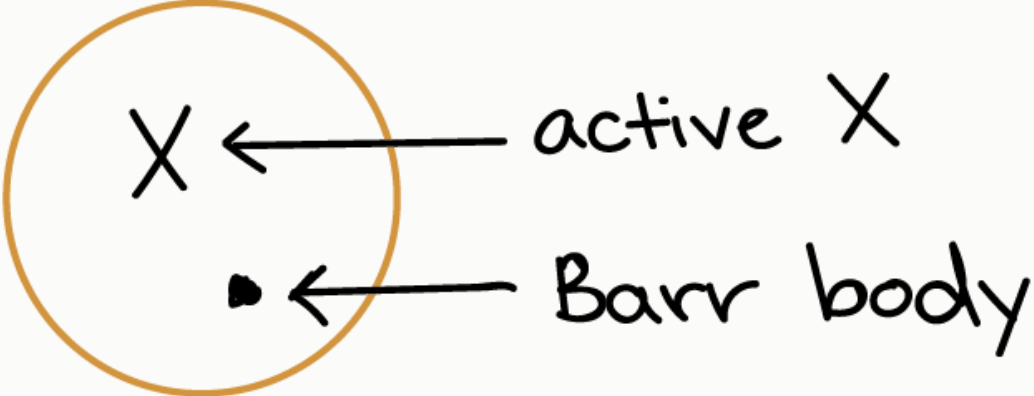
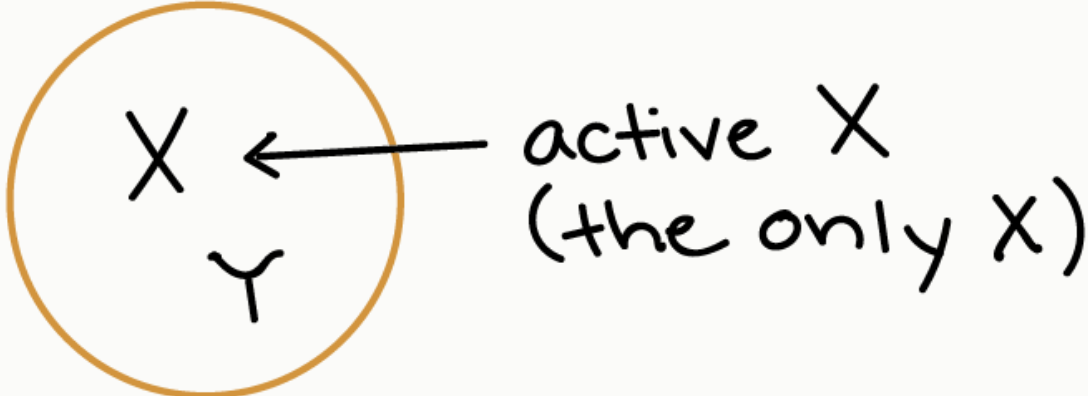
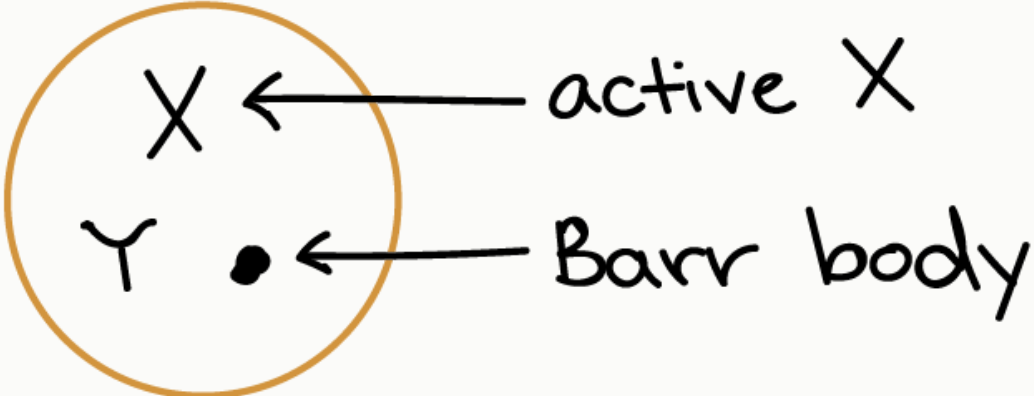
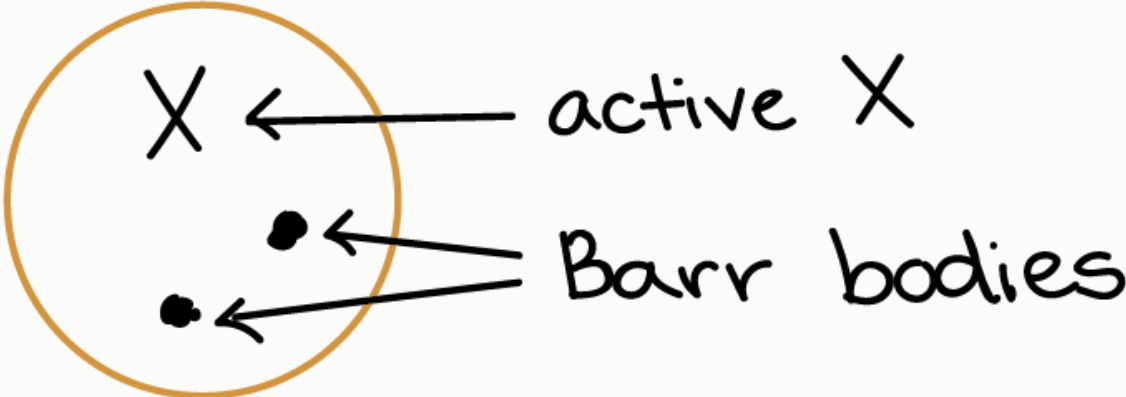
47, XXX
48, XXXY
($N - 1 = 2$)



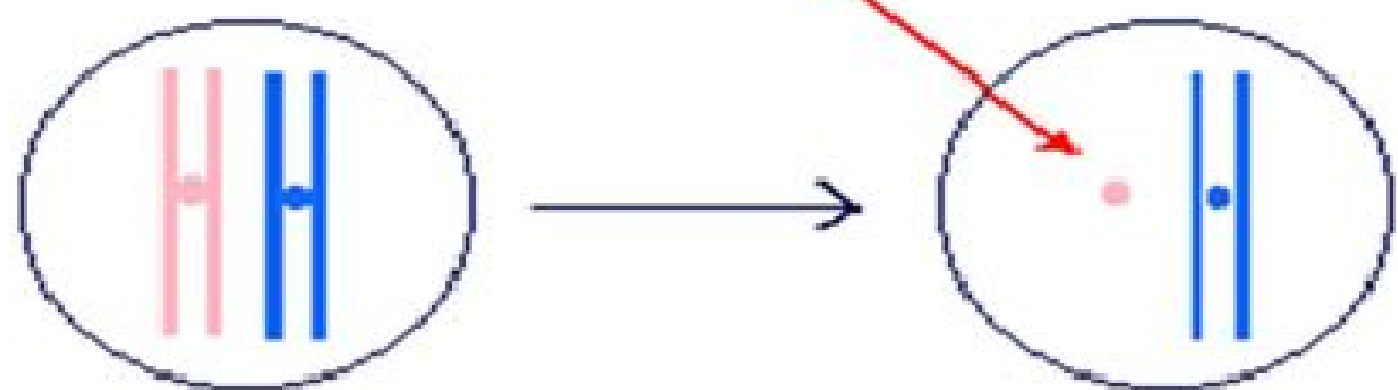
48, XXXX
49, XXXXY
($N - 1 = 3$)

46, XX	1 Barr body
45, X	0 Barr body
47, XXY	1 Barr body
47, XXX	2 Barr bodies
48, XXXX	3 Barr bodies

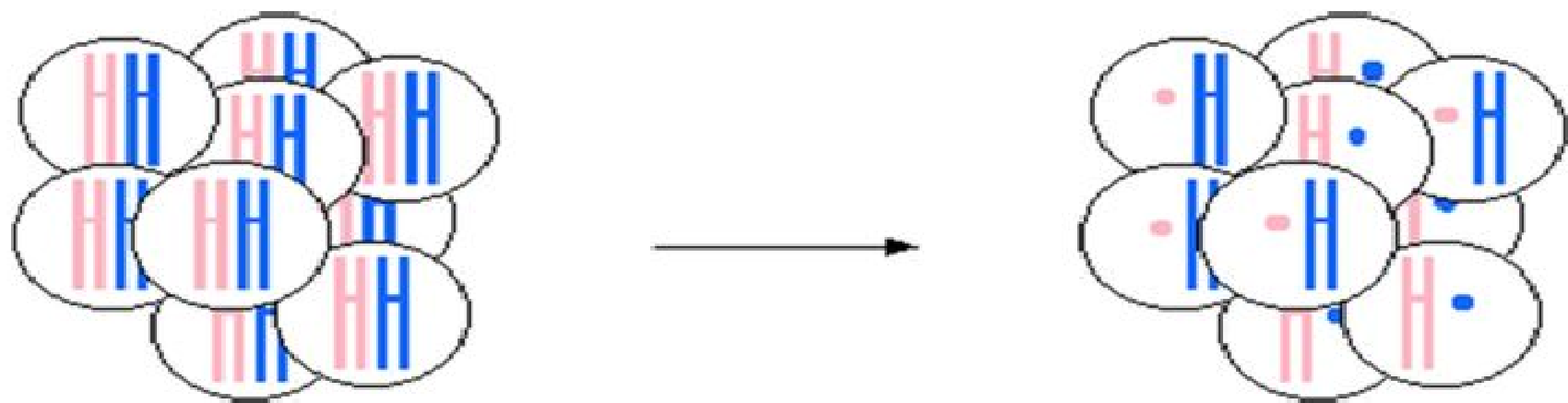
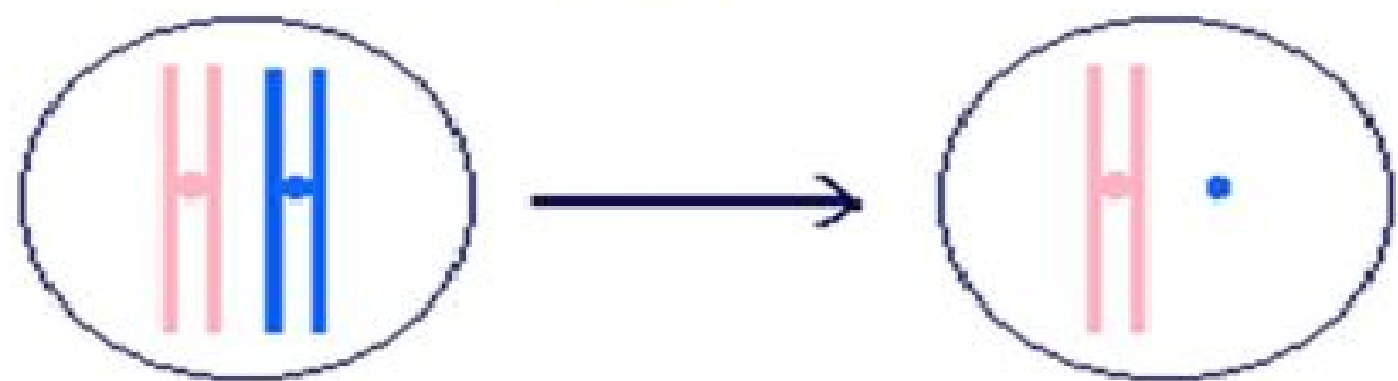
Barr bodies = $N - 1$ ($N = \# \text{ X cs. present}$)

XX female	 Diagram of a female cell (XX) showing one active X chromosome and one Barr body. The active X is represented by a large 'X' and the Barr body by a small black dot, both enclosed in a circle. Arrows point from the labels 'active X' and 'Barr body' to their respective structures.
XY male	 Diagram of a male cell (XY) showing one active X chromosome. The active X is represented by a large 'X' and the Y chromosome by a small 'Y', both enclosed in a circle. An arrow points from the label 'active X (the only X)' to the active X chromosome.
XXY male (Klinefelter)	 Diagram of a Klinefelter male cell (XXY) showing one active X chromosome and one Barr body. The active X is represented by a large 'X' and the Barr body by a small black dot, both enclosed in a circle. Arrows point from the labels 'active X' and 'Barr body' to their respective structures.
XXX female (triple X)	 Diagram of a triple X female cell (XXX) showing one active X chromosome and two Barr bodies. The active X is represented by a large 'X' and the two Barr bodies by two small black dots, all enclosed in a circle. Arrows point from the labels 'active X' and 'Barr bodies' to their respective structures.

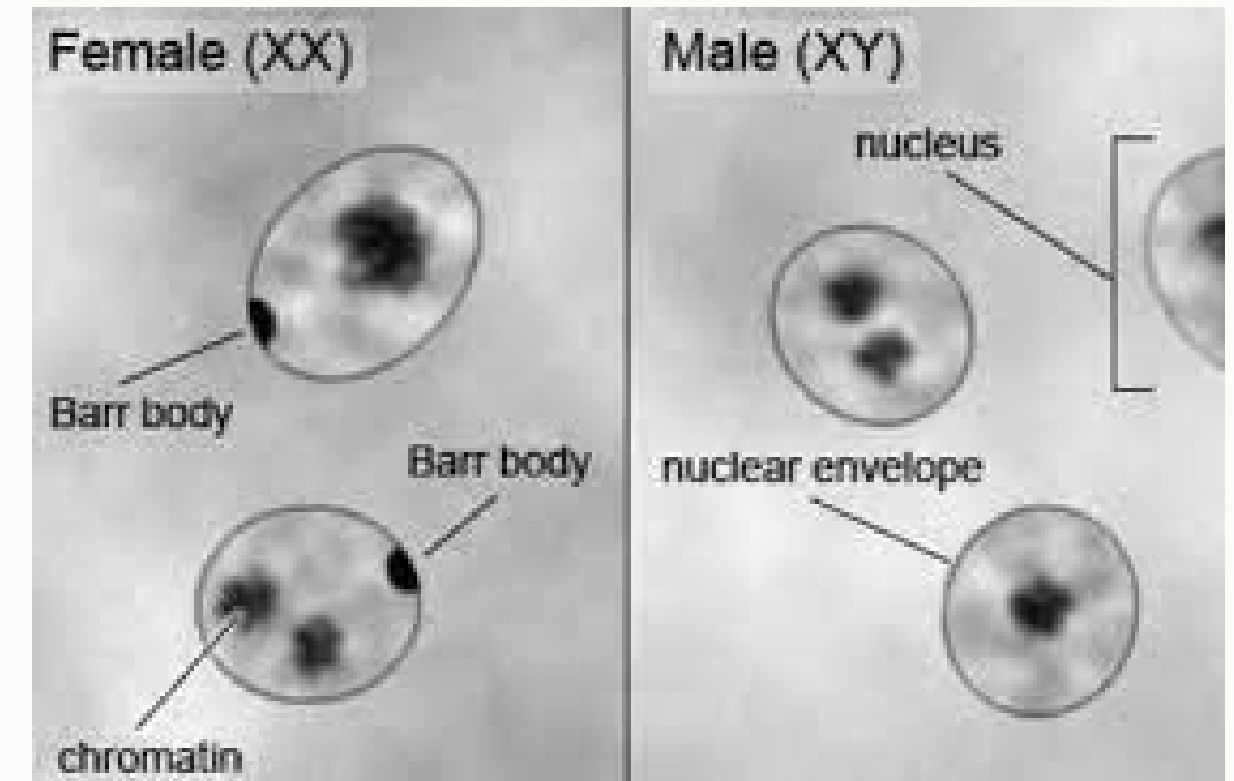
Barr body



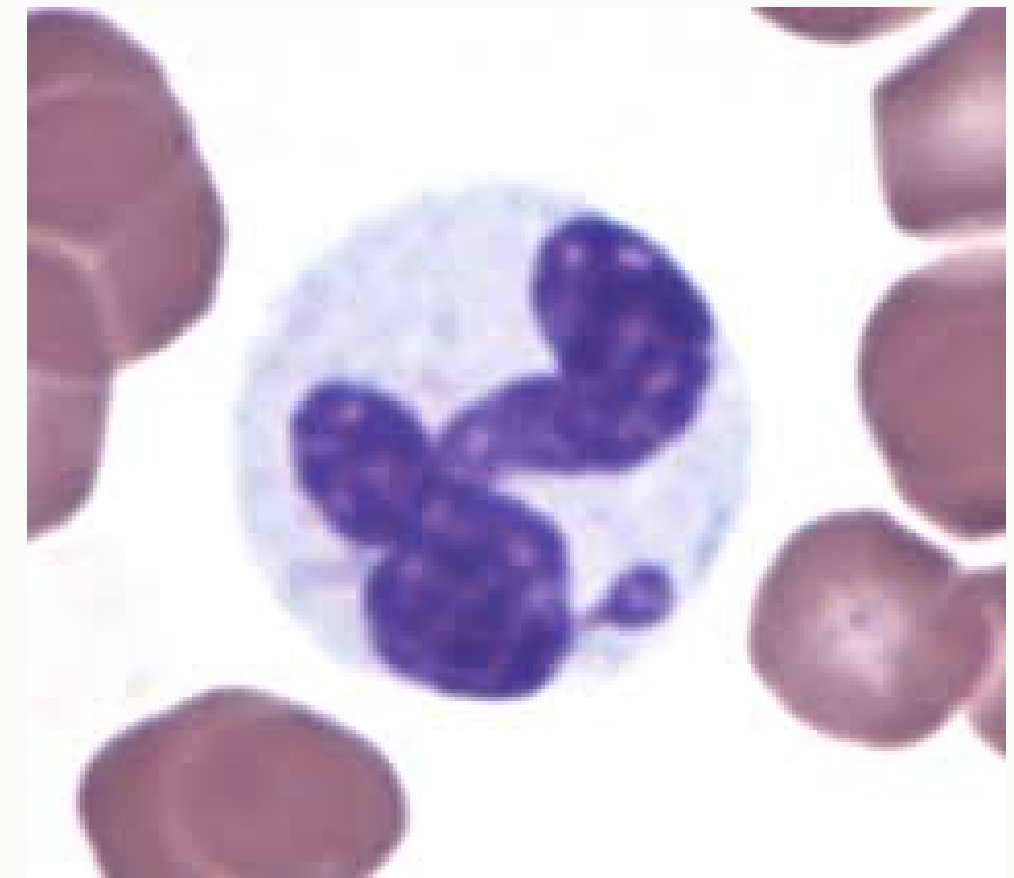
OR



- **Barr Body presents in all somatic cells as a condensed dark mass next to the periphery of the nuclear envelope**

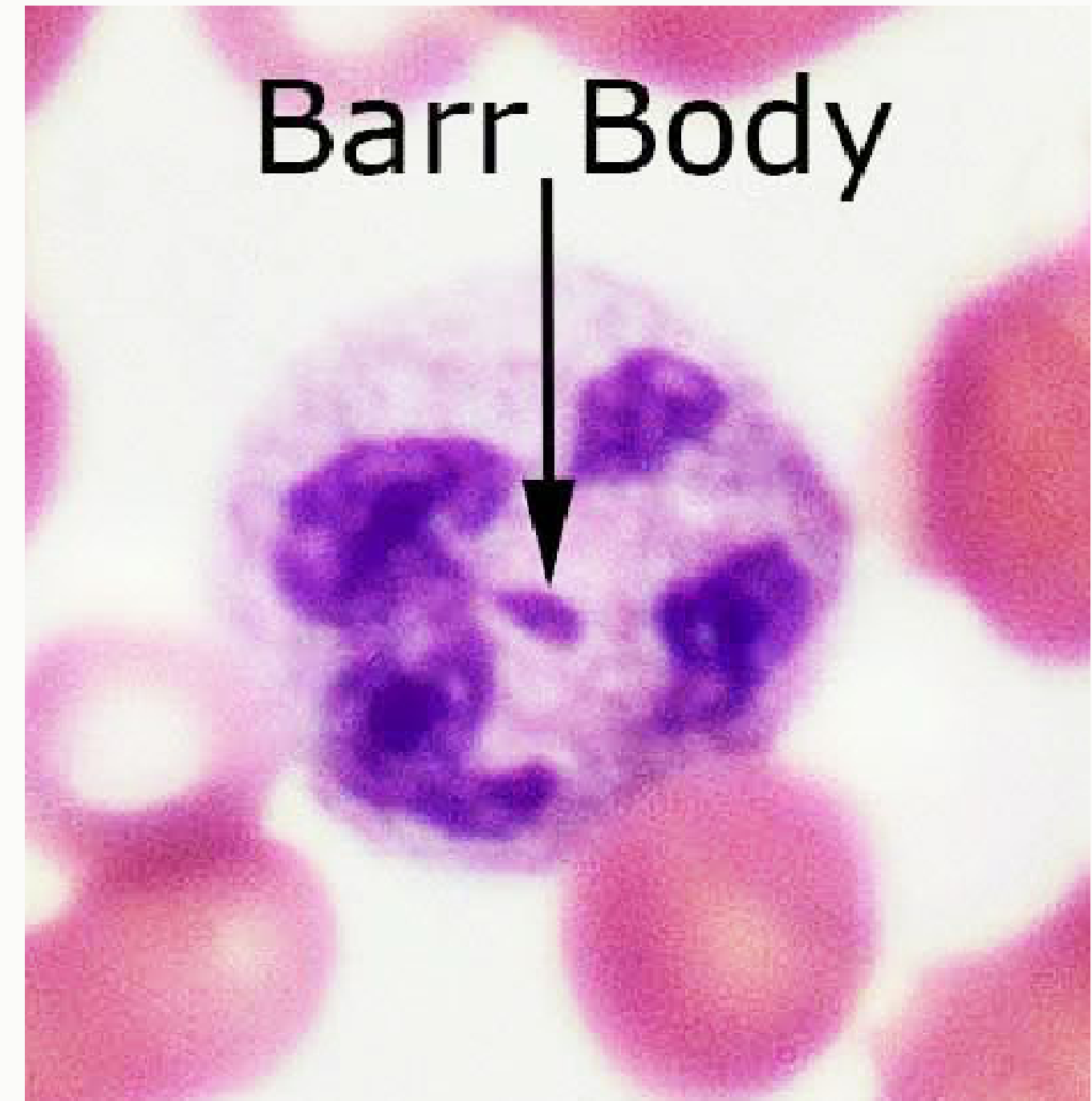


- **Barr Body presents in neutrophils as small rod attached to the nuclear lobe by a single narrow stalk (Drumstick)**





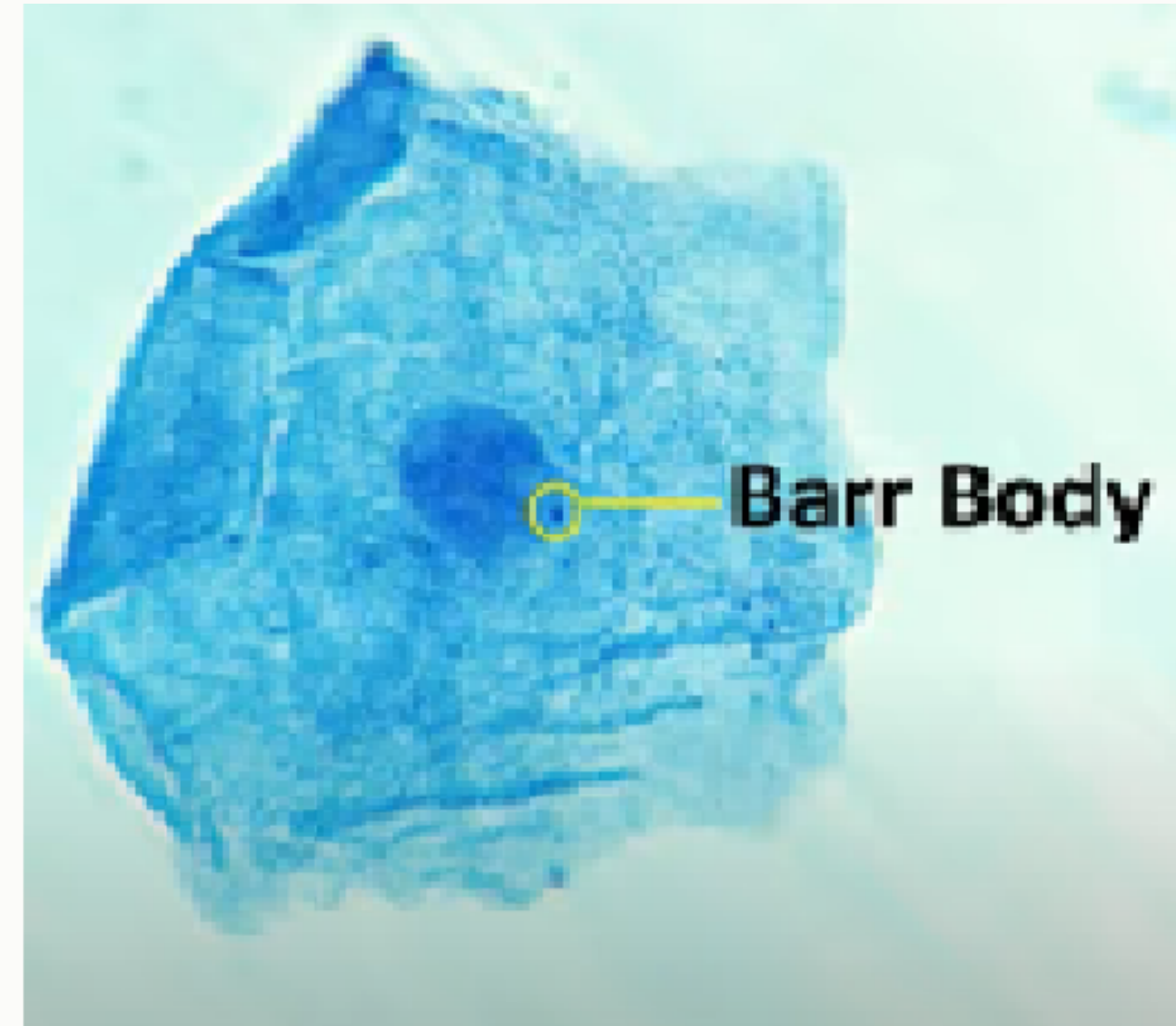
Barr Body in Somatic Cell



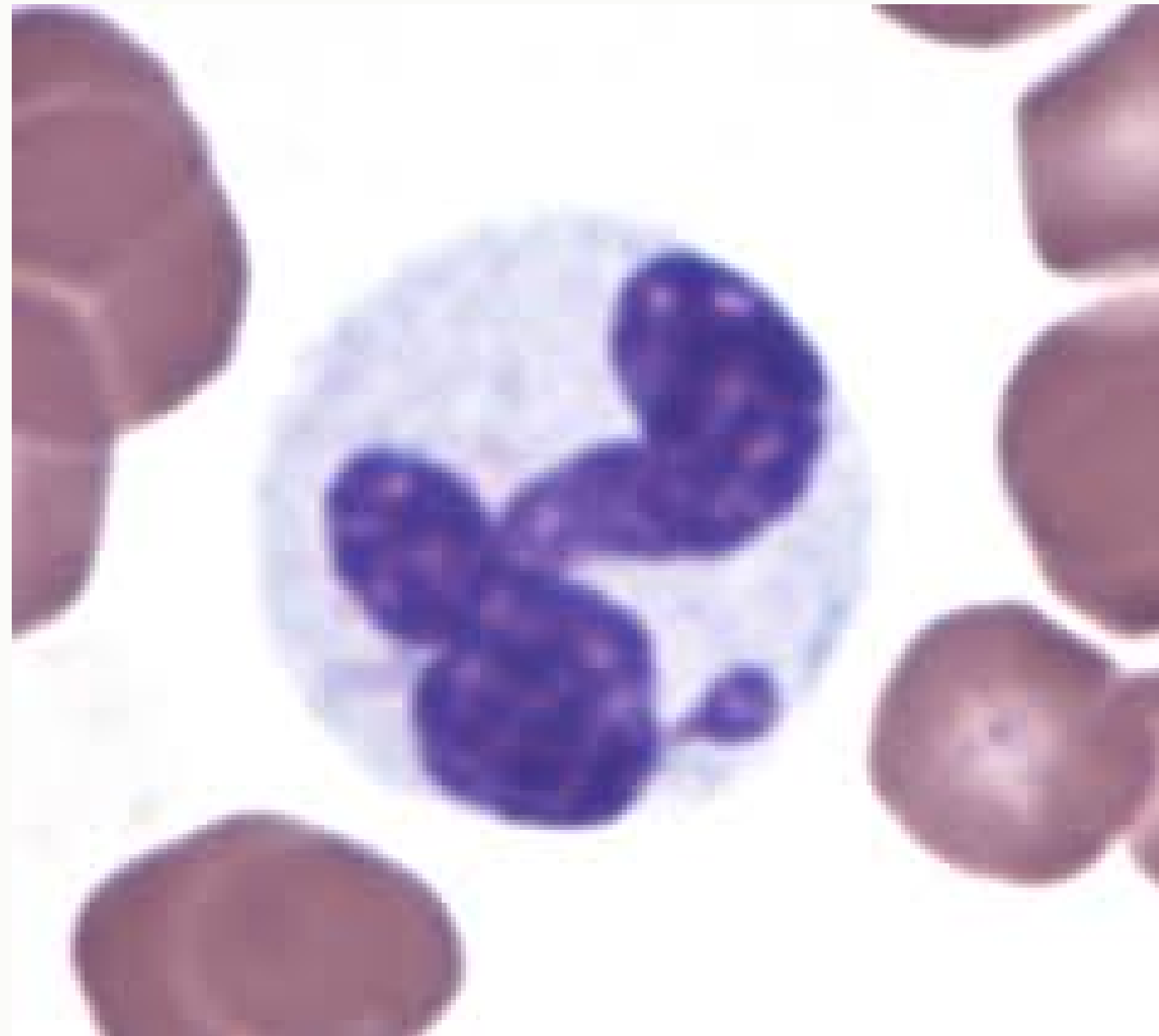
Drumstick in Neutrophile

Methodology

1. Obtain a clean glass microscope slide.
2. Place a drop of water in the center of the slide.
3. Gently scrape the inside of your cheek with the flat edge of a toothpick.
4. Spread cells in the drop of water.
5. Place cover slip on top of specimen. Use correct technique to prevent air bubbles.
7. Apply one drop of methylene blue along one side of the cover slip.
8. Along the opposite edge of the cover slip, gently touch with a piece of paper towel. This will draw the methylene blue under the cover slip to stain the cells.



Preparation of the Blood Smear: Drumsticks

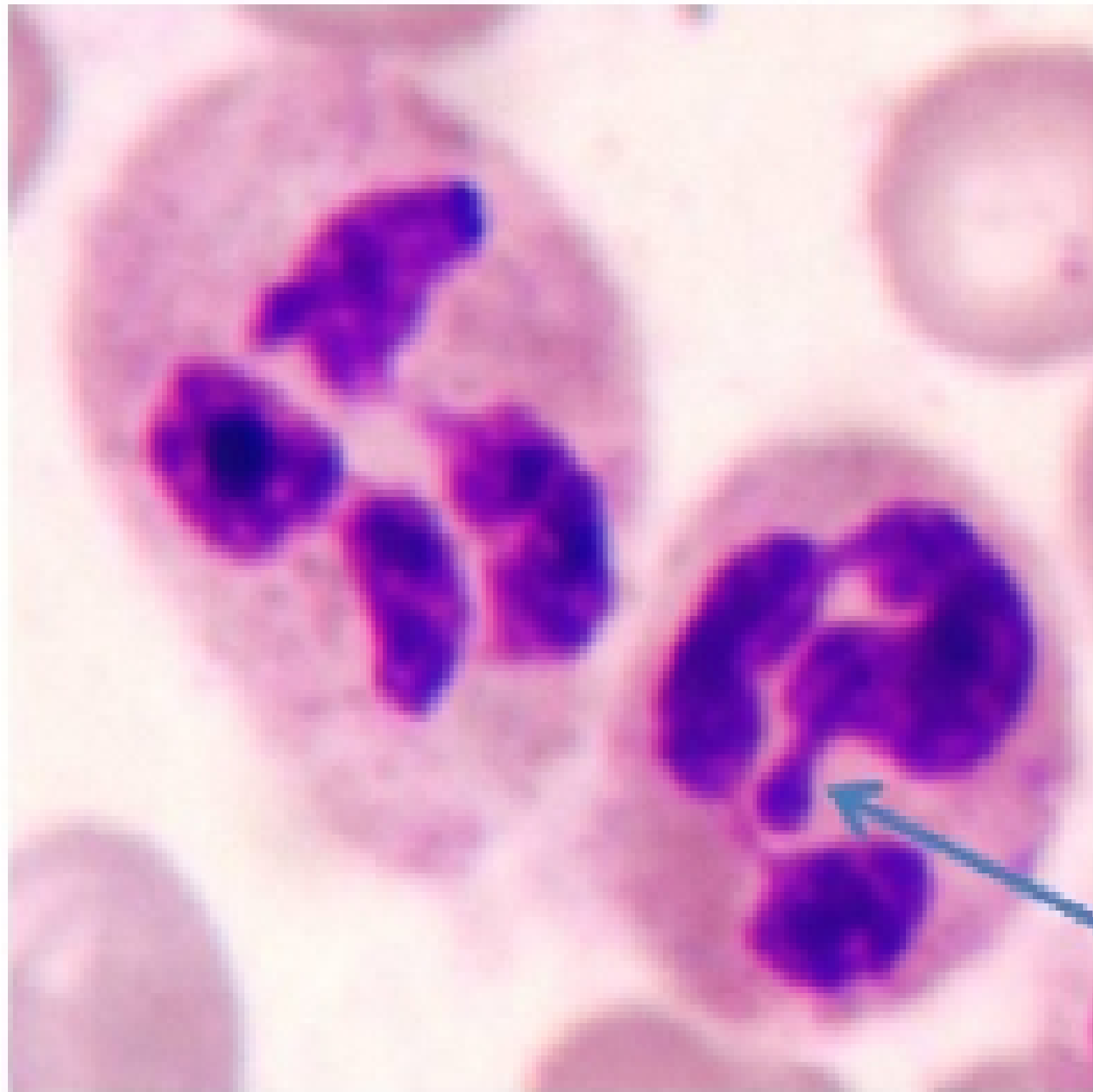


NEUTROPHIL

- **The neutrophil are the more common leukocytes. They have a diameter of 12-15 μm .**
- **You can recognize them as their nucleus is divided into 2 - 5 lobes connected by a fine nuclear strand or filament.**
- **In the nucleus of the neutrophil of cells from females, you may see an appendage like a little drumstick (Barr body).**
- **It is the second X chromosome, inactivated.**

Methodology

- 1. Place a small drop of blood near an end of a slide.**
- 2. Bring the edge of another slide in contact with the drop and allow the drop to bank evenly behind the spreader.**
- 3. The angle between the two slides has to be 30-40 degrees.**
- 4. Now, push to the left in a smooth, quick motion.**
- 5. Air drying**
- 6. dipping the smear in a vessel containing 95% ethyl or methyl alcohol for 3-5 min.**
- 7. Put on the slide a drop of stain until it is fully covered, Stain for about 16 minutes, renewing the stain about four times.**
- 8. Rinse the slide with distilled water at room temperature.**
- 9. Drain off the water and leave the slide to dry.**



Neutrophil
Drumstick